

antVacDB — what we extract from each paper

Pydantic schema **v2.7.1** · 13 entity types · 26 controlled vocabularies. One paper → one `ExtractedPaper` record.

How to read this

- Every paper becomes one **ExtractedPaper**, which holds lists of the entities below (patients, peptides, epitopes, evidence rows, survival outcomes...).
- **Two peptide levels:** an **ImmunizingPeptide** is the long peptide actually given; a **MinimalEpitope** is the short MHC-binding stretch within it (one long peptide can yield several).
- **Nothing is invented.** A value not stated in the paper is left empty. Numbers read off a figure are kept lossless (the raw reading is preserved, flagged low-confidence, never a clean number).
- **Controlled vocabularies** (bottom) are the only allowed values for the categorical fields — they keep extractions comparable across papers.

[ExtractedPaper](#) · [ExtractedPatient](#) · [ImmunizingPeptide](#) · [MinimalEpitope](#) · [ExtractedEvidence](#) · [ExtractedPeptidePool](#) · [SurvivalOutcome](#) · [PreclinicalEfficacy](#) · [ConcomitantImmunosuppression](#) · [CuratorNote](#) · [Measurement](#) · [ResponseMagnitude](#) · [Provenance](#) · [Controlled vocabularies](#)

Entities

ExtractedPaper 17 fields

Top-level container: one paper, all extracted entities.

field	type	presence	notes
<code>pmid</code>	str	required	
<code>doi</code>	str format <code>^10\.\d{4,9}/[-._;()/:A-Z0-9a-z]+\$</code> (optional)	optional	
<code>pmcid</code>	str format <code>^PMC\d+\$</code> (optional)	optional	
<code>nct_id</code>	str format <code>^NCT\d{8}\$</code> (optional)	optional	
<code>journal</code>	str	required	
<code>year</code>	int	required	
<code>title</code>	str	required	

field	type	presence	notes
cohort_size	int	required	
n_enrolled	int <i>(optional)</i>	<i>optional</i>	
indication_summary	str	required	
patients	list of ExtractedPatient	<i>optional</i>	
immunizing_peptides	list of ImmunizingPeptide	<i>optional</i>	
epitopes	list of MinimalEpitope	<i>optional</i>	
pools	list of ExtractedPeptidePool	<i>optional</i>	
evidence	list of ExtractedEvidence	<i>optional</i>	
survival_outcomes	list of SurvivalOutcome	<i>optional</i>	
curator_notes	list of CuratorNote	<i>optional</i>	

ExtractedPatient 20 fields

One trial subject as reported in the paper.

`paper\_local\_id` is whatever the paper itself uses ('P1', 'Patient 3', 'pt07'). The fully-qualified `patient\_id` written to the DB is `pmid:<PMID>:<paper\_local\_id>` — assembled at ingest time, not by the LLM.

Patient identifiers must be paper-local labels — never names, initials, or medical record numbers. The Phase D validators reject fields that look name-like.

field	type	presence	notes
quoted_text	str	required	
section_ref	str	required	
provenance	list of Provenance	<i>optional</i>	
confidence	int <i>(optional)</i>	<i>optional</i>	
needs_review	bool	<i>optional</i> default False	
paper_local_id	str	required	
indication	str	required	
species	one of: human · mouse · rat · non_human_primate · other	<i>optional</i> default	

field	type	presence	notes
		human	
model_system	str ≥1 chars <i>(optional)</i>	<i>optional</i>	
vaccine_platform	one of: dna · rna · synthetic_long_peptide · short_peptide · dendritic_cell · viral_vector · other · unspecified	<i>optional</i> default unspecified	
vaccine_platform_detail	str ≤200 chars <i>(optional)</i>	<i>optional</i>	
hla_alleles	list of str format <code>^(HLA-[A-Z][A-Z0-9]**\d{2,3}:\d{2,3} HLA-D[PQ]A1*\d{2,3}:\d{2,3}/D[PQ]B1*\d{2,3}:\d{2,3} H-2[KDLQ][a-z]\d? H-2I[AE][a-z] I-[AE][a-z])\$</code>	<i>optional</i>	
n_peptides_synthesized	int	required	
n_peptides_administered	int <i>(optional)</i>	<i>optional</i>	
n_peptides_immunogenic	int	required	
immunogenicity_assessed	bool <i>(optional)</i>	<i>optional</i>	
clinical_outcome	one of: CR · PR · SD · PD · not_reported <i>(optional)</i>	<i>optional</i>	
best_response	one of: CR · PR · SD · PD · not_reported <i>(optional)</i>	<i>optional</i>	
immunosuppression	list of ConcomitantImmunosuppression	<i>optional</i>	
preclinical_efficacy	list of PreclinicalEfficacy	<i>optional</i>	

ImmunizingPeptide 15 fields

One long peptide actually synthesized/administered — the *pool member*.

This is the entity `ExtractedPeptidePool.member\_peptide\_ids` points at and `MinimalEpitope.parent\_peptide\_ids` points back to. For personalized neoantigen vaccines the peptide belongs to one patient; `patient\_paper\_id` records that (None for shared/off-the-shelf antigens). When set, it powers the per-patient count reconciliation in `ExtractedPaper` (v2 P8).

field	type	presence	notes
quoted_text	str	required	
section_ref	str	required	
provenance	list of Provenance	<i>optional</i>	

field	type	presence	notes
confidence	int <i>(optional)</i>	<i>optional</i>	
needs_review	bool	<i>optional</i> default False	
paper_local_id	str	required	
sequence	str	required	
gene_symbol	str ≥1 chars <i>(optional)</i>	<i>optional</i>	
uniprot_id	str format <code>^([0PQ] [0-9] [A-Z0-9] {3} [0-9] [A-NR-Z] [0-9] ([A-Z] [A-Z0-9] {2} [0-9]) {1,2})\$</code> <i>(optional)</i>	<i>optional</i>	
is_neoantigen	bool	required	
mutation	str ≤64 chars <i>(optional)</i>	<i>optional</i>	
variant_type	one of: <code>missense</code> · <code>frameshift</code> · <code>inframe_indel</code> · <code>stop_loss</code> · <code>splice</code> · <code>fusion</code> · <code>other</code> <i>(optional)</i>	<i>optional</i>	
protein_change	str format <code>^p\.\ (? [A-Za-z0-9_*=>] + \) ?\$</code> · ≤64 chars <i>(optional)</i>	<i>optional</i>	
hla_allele	str format <code>^(HLA-[A-Z] [A-Z0-9] **\d{2,3}:\d{2,3} HLA-D[PQ]A 1*\d{2,3}:\d{2,3}/D[PQ]B1*\d{2,3}:\d{2,3} H-2[KDLQ] [a-z] \d? H-2I[AE] [a-z] I-[AE] [a-z])\$</code> <i>(optional)</i>	<i>optional</i>	
patient_paper_id	str ≥1 chars <i>(optional)</i>	<i>optional</i>	

MinimalEpitope 19 fields

A minimal epitope WITHIN one or more immunizing peptides (v2 P1/P2/P3).

`parent\_peptide\_ids` is MANY-TO-MANY: overlapping/tiling long peptides can share an epitope (confirmed in Keskin Supp Table 5 — e.g. EPT2B sits in IMP02/03/04). Empty list = the paper reported the epitope without resolving which long peptide it came from (preserved honestly, not invented).

`mhc\_class` is mandatory because it governs which other fields are even meaningful: a class-I epitope (8–14mer) carries a NetMHCpan nM affinity or %rank; a class-II / CD4 epitope does NOT carry a class-I affinity and is typically restricted by HLA-DR/DQ/DP (often un-typed in the paper).

Q9 (unchanged): in-silico prediction metadata are PREDICTIONS, never evidence — they must NEVER seed an ExtractedEvidence row. If the only thing known about an epitope is its predicted affinity, it has zero evidence rows (predicted/T0 tier), not a 'presented'/'positive' row.

field	type	presence	notes
quoted_text	str	required	
section_ref	str	required	
provenance	list of Provenance	<i>optional</i>	
confidence	int (<i>optional</i>)	<i>optional</i>	
needs_review	bool	<i>optional</i> default False	
paper_local_id	str	required	
sequence	str	required	
gene_symbol	str ≥1 chars (<i>optional</i>)	<i>optional</i>	
uniprot_id	str format <code>^([OPQ] [0-9] [A-Z0-9]{3} [0-9] [A-NR-Z] [0-9] ([A-Z] [A-Z0-9]{2} [0-9]) {1,2})\$</code> (<i>optional</i>)	<i>optional</i>	
is_neoantigen	bool	required	
mutation	str ≤64 chars (<i>optional</i>)	<i>optional</i>	
variant_type	one of: missense · frameshift · inframe_indel · stop_loss · splice · fusion · other (<i>optional</i>)	<i>optional</i>	
protein_change	str format <code>^p\.\.(? [A-Za-z0-9_*=>]+\)?\$</code> · ≤64 chars (<i>optional</i>)	<i>optional</i>	
hla_allele	str format <code>^(HLA-[A-Z] [A-Z0-9]**\d{2,3}:\d{2,3} HLA-D[PQ]A1*\d{2,3}:\d{2,3}/D[PQ]B1*\d{2,3}:\d{2,3} H-2[KDLQ][a-z]\d? H-2I[AE][a-z] I-[AE][a-z])\$</code> (<i>optional</i>)	<i>optional</i>	
parent_peptide_ids	list of str ≥1 chars	<i>optional</i>	
mhc_class	one of: I · II	required	
wild_type_sequence	str (<i>optional</i>)	<i>optional</i>	
predicted_affinity	Measurement (<i>optional</i>)	<i>optional</i>	
wild_type_affinity	Measurement (<i>optional</i>)	<i>optional</i>	

ExtractedEvidence 22 fields

One evidence row linking a patient × target to an assay outcome.

The (assay, stimulation, antigen_format, timepoint) tuple identifies a distinct *measurement* — two timepoints of the same assay on the same peptide×patient are two evidence rows, not one. See the v2 design doc.

field	type	presence	notes
quoted_text	str	required	
section_ref	str	required	
provenance	list of Provenance	<i>optional</i>	
confidence	int (<i>optional</i>)	<i>optional</i>	
needs_review	bool	<i>optional</i> default False	
patient_paper_id	str	required	
target_kind	one of: immunizing_peptide · epitope · pool	<i>optional</i> default epitope	
immunizing_peptide_paper_id	str ≥1 chars (<i>optional</i>)	<i>optional</i>	
epitope_paper_id	str ≥1 chars (<i>optional</i>)	<i>optional</i>	
pool_paper_id	str ≥1 chars (<i>optional</i>)	<i>optional</i>	
hla_allele	str format <code>^(HLA-[A-Z][A-Z0-9]*\d{2,3}:\d{2,3} HLA-D[PQ]A1*\d{2,3}:\d{2,3}/D[PQ]B1*\d{2,3}:\d{2,3} H-2[KDLQ][a-z]\d? H-2I[AE][a-z] I-[AE][a-z])\$</code> (<i>optional</i>)	<i>optional</i>	
hla_source	str ≤64 chars (<i>optional</i>)	<i>optional</i>	
assay	one of: elispot · tetramer · ics · flow_cytometry · ms · binding · tcr_reporter · vaccine_response · clinical_response	required	
outcome	one of: immunogenic · not_immunogenic · pre_existing · not_assessed · positive · negative · presented	required	
vaccine_induced	bool (<i>optional</i>)	<i>optional</i>	
mutation_specific	bool (<i>optional</i>)	<i>optional</i>	
assay_stimulation	one of: ex_vivo · in_vitro_expanded · unspecified (<i>optional</i>)	<i>optional</i>	
assay_antigen_format	one of: peptide_pulsed · minigene · autologous_tumor · tetramer · unspecified (<i>optional</i>)	<i>optional</i>	
timepoint_phase	one of: pre_vaccine · on_treatment · post_vaccine · memory · unspecified (<i>optional</i>)	<i>optional</i>	
timepoint_label	str ≤64 chars (<i>optional</i>)	<i>optional</i>	

field	type	presence	notes
assay_detail	str ≤200 chars (<i>optional</i>)	<i>optional</i>	
magnitude	ResponseMagnitude (<i>optional</i>)	<i>optional</i>	

ExtractedPeptidePool 8 fields

A named immunizing pool within a paper (e.g. Keskin's pool 'A'..'D').

Pools are per-patient here. A pool-target ExtractedEvidence row references `paper\_local\_id`. peptide <-> pool is MANY-TO-MANY (a peptide can sit in multiple pools — confirmed in Keskin Supp Table 5): `member\_peptide\_ids` holds every peptide in the pool; the DB join table is built from it.

PATCH (v2 P1): `member\_peptide\_ids` now references `ImmunizingPeptide` paper_local_ids (the long, injected peptides) — NOT minimal epitopes. A pool physically contains long peptides; epitopes live inside those.

field	type	presence	notes
quoted_text	str	required	
section_ref	str	required	
provenance	list of Provenance	<i>optional</i>	
confidence	int (<i>optional</i>)	<i>optional</i>	
needs_review	bool	<i>optional</i> default False	
paper_local_id	str	required	
patient_paper_id	str	required	
member_peptide_ids	list of str ≥1 chars	required	

SurvivalOutcome 16 fields

One reported time-to-event outcome — the human analog of PreclinicalEfficacy, for endpoints RANO cannot express (RFS/OS/...).

Modeled as ONE record per (endpoint x arm). A two-arm comparison is two records: the experimental arm carries the comparison stats (hazard_ratio, p_value, comparator_label); the reference arm is a plain median record.

Lives at the PAPER level — a between-group comparison spans patient groups, not a single patient. Per-patient IPD (if a paper ever reports it) would be

a separate per-patient addition; this captures what trials actually print.

Rojas RFS-by-response:

- arm='vaccine responders', n=8, not_reached=True, comparator='non-responders', hazard_ratio=0.08, hr_ci='0.01-0.4', p_value=0.003, test='log-rank', stratifier='vaccine response', median_followup_value=18.0
- arm='non-responders', n=8, median_value=13.4 (reference arm)

field	type	presence	notes
endpoint	one of: <code>rfs</code> · <code>os</code> · <code>landmark_rfs</code> · <code>dfs</code> · <code>pfs</code> · <code>efs</code> · <code>ttr</code> · <code>other</code>	required	
arm_label	str ≥1 chars (<i>optional</i>)	<i>optional</i>	
n_patients	int (<i>optional</i>)	<i>optional</i>	
median_value	float (<i>optional</i>)	<i>optional</i>	
time_unit	one of: <code>months</code> · <code>weeks</code> · <code>days</code> · <code>years</code>	<i>optional</i> default months	
not_reached	bool	<i>optional</i> default False	
events	int (<i>optional</i>)	<i>optional</i>	
median_followup_value	float (<i>optional</i>)	<i>optional</i>	
comparator_label	str ≥1 chars (<i>optional</i>)	<i>optional</i>	
hazard_ratio	float (<i>optional</i>)	<i>optional</i>	
hr_ci	str ≤40 chars (<i>optional</i>)	<i>optional</i>	
p_value	float (<i>optional</i>)	<i>optional</i>	
statistical_test	str ≤48 chars (<i>optional</i>)	<i>optional</i>	
stratifier	str ≤80 chars (<i>optional</i>)	<i>optional</i>	
note	str ≤300 chars (<i>optional</i>)	<i>optional</i>	
source	Provenance (<i>optional</i>)	<i>optional</i>	

PreclinicalEfficacy 11 fields

One antitumor-efficacy experiment arm, as reported — the preclinical analog of `clinical\_outcome`. Recorded as a FACT (arm -> readout -> result), not the paper's causal story.

A list because efficacy depends on the treatment ARM. Li et al. 2021, E0771:

- vaccine alone -> readout='tumor_growth', result='no_effect',

combination='monotherapy'

- vaccine + anti-PD-L1 -> result='growth_inhibition',

combination='plus_checkpoint_inhibitor'

4T1.2: vaccine alone -> 'partial_inhibition' (note: model is anti-PD-L1 resistant; combination not yet tested). For HUMAN cohorts the primary outcome belongs in `clinical\_outcome`/'best_response`; leave this empty.

field	type	presence	notes
readout	one of: tumor_growth · survival · tumor_free_fraction · metastasis · other	required	
result	one of: no_effect · partial_inhibition · growth_inhibition · tumor_regression · complete_response · prolonged_survival · not_reported	required	
setting	one of: prophylactic · therapeutic · unspecified	optional default unspecified	
combination	one of: monotherapy · plus_checkpoint_inhibitor · plus_chemotherapy · plus_radiotherapy · plus_other · unspecified	optional default monotherapy	
combination_detail	str ≤120 chars (optional)	optional	
comparator	str ≤120 chars (optional)	optional	
n_animals	int (optional)	optional	
timepoint	str ≤64 chars (optional)	optional	
statistic	str ≤120 chars (optional)	optional	
note	str ≤300 chars (optional)	optional	
source	Provenance (optional)	optional	

ConcomitantImmunosuppression 5 fields

One immunosuppressive exposure during the vaccine window, recorded as a FACT — never the paper's interpretation of its effect.

Keskin Supp Table 4a: 6/8 patients on dexamethasone 2-4 mg/day during vaccine priming -> agent_class='corticosteroid', agent='dexamethasone', dose='2-4 mg/day', timing='during_priming'. The schema does NOT store "this caused the response to fail" — that correlation is derived across the corpus (steroid-during-priming vs n_peptides_immunogenic == 0), not asserted.

field	type	presence	notes
agent_class	one of: corticosteroid · chemotherapy · calcineurin_inhibitor · mtor_inhibitor · antimetabolite · anti_tnf · other_immunosuppressant · unspecified	required	
agent	str ≥1 chars (<i>optional</i>)	<i>optional</i>	
dose	str ≤64 chars (<i>optional</i>)	<i>optional</i>	
timing	one of: pre_vaccine · during_priming · during_boost · throughout · post_vaccine · unspecified	<i>optional</i> default unspecified	
source	Provenance (<i>optional</i>)	<i>optional</i>	

CuratorNote 6 fields

One piece of curatorial commentary about the paper. Optional, paper-level, and rendered verbatim — see the anti-hallucination note above. Empty for every record unless a human/scoped pass deliberately adds one.

field	type	presence	notes
kind	one of: challenge · decision · caveat · highlight	required	
text	str	required	
refs	list of str ≥1 chars	<i>optional</i>	
confidence	int (<i>optional</i>)	<i>optional</i>	
needs_review	bool	<i>optional</i> default True	
source	Provenance (<i>optional</i>)	<i>optional</i>	

Measurement 8 fields

A single quantitative measurement or prediction — unit-explicit and lossless.

Replaces bare ``\*\_nm: float`` fields. Three reasons:

1. UNIT IS DATA. nM and NetMHCpan %rank are different scales that live in the SAME source column (Keskin Supp Table 5 mixes them — ARHGAP35, a validated hit, is reported as %rank). Carrying the unit on the value makes "0.80 (%rank)" un-confusable with 0.80 nM, which collapses the

former two-field P3 split into one well-typed object.

2. LOSSLESS. ``raw`` keeps the verbatim source token regardless of whether parsing succeeded, so a future/better parser can re-derive ``value`` and retroactively upgrade already-stored records. ``value`` may be None (parse failed) as long as ``raw`` survives — a measurement is never silently dropped or coerced.
3. PROVENANCED CLAIM. method + version + tier + source let contradictory claims about the same quantity (NetMHCpan v2.4 vs v4.1 vs the paper's own printed value) coexist side by side instead of overwriting.

field	type	presence	notes
value	float (<i>optional</i>)	<i>optional</i>	
unit	one of: nM · pct_rank · unknown	required	
raw	str len 1–2000 (<i>optional</i>)	<i>optional</i>	
method	str ≥1 chars (<i>optional</i>)	<i>optional</i>	
method_version	str ≤32 chars (<i>optional</i>)	<i>optional</i>	
tier	one of: predicted · reported · validated	<i>optional</i> default predicted	
source	Provenance (<i>optional</i>)	<i>optional</i>	
confidence	int (<i>optional</i>)	<i>optional</i>	

ResponseMagnitude 9 fields

How big a measured T-cell response was — the structured home for the ELISpot / ICS / tetramer magnitude that previously fell into `ExtractedEvidence.assay\_detail` free text. Attaches to ONE evidence row (one assay x timepoint).

Any lane may be filled: a parsed `value` + `unit` (e.g. 2459 sfc_per_1e6); an ordinal `grade` (murine -/+ /+++ /++++ /+++++ -> negative/low/moderate/high/very_high); or — losslessly — only `raw`, when the paper gives a magnitude that can't be parsed or mapped (a per-patient SFC set without gene labels). At least one of value / grade / raw must be present (never an empty size).

field	type	presence	notes
value	float (<i>optional</i>)	<i>optional</i>	
unit	one of: sfc_per_1e6 · percent_of_parent · stimulation_index · unknown	<i>optional</i> default unknown	

field	type	presence	notes
grade	one of: <code>negative</code> · <code>low</code> · <code>moderate</code> · <code>high</code> · <code>very_high</code> (<i>optional</i>)	<i>optional</i>	
raw	str len 1–2000 (<i>optional</i>)	<i>optional</i>	
background_subtracted	bool (<i>optional</i>)	<i>optional</i>	
denominator	str ≤48 chars (<i>optional</i>)	<i>optional</i>	
tier	one of: <code>predicted</code> · <code>reported</code> · <code>validated</code>	<i>optional</i> default reported	
source	Provenance (<i>optional</i>)	<i>optional</i>	
confidence	int (<i>optional</i>)	<i>optional</i>	

Provenance 3 fields

One structured source pointer for an extracted fact.

The legacy ``quoted_text`` + ``section_ref`` on each row remain the required, human-readable anchor. ``Provenance`` adds a **typed, repeatable** pointer so a fact backed by several places (e.g. Keskin dexamethasone in BOTH Supp Table 4a and the Results prose) records all of them, and so Phase D can weight a table-sourced value differently from a figure-read one.

field	type	presence	notes
kind	one of: <code>table</code> · <code>figure</code> · <code>prose</code> · <code>lookup</code>	required	
locator	str	required	
quoted_text	str len 1–2000 (<i>optional</i>)	<i>optional</i>	

Controlled vocabularies

The allowed values for each categorical axis (the extractor cannot use anything else).

ASSAYS

`elispot` `tetramer` `ics` `flow_cytometry` `ms` `binding`
`tcr_reporter` `vaccine_response` `clinical_response`

ASSAY_ANTIGEN_FORMAT

`peptide_pulsed` `minigene` `autologous_tumor` `tetramer`
`unspecified`

ASSAY_STIMULATION

`ex_vivo` `in_vitro_expanded` `unspecified`

COMBINATION_CLASSES	monotherapy	plus_checkpoint_inhibitor	plus_chemotherapy	plus_radiotherapy	plus_other	unspecified
CURATOR_NOTE_KINDS	challenge	decision	caveat	highlight		
EFFICACY_READOUTS	tumor_growth	survival	tumor_free_fraction	metastasis	other	
EFFICACY_RESULTS	no_effect	partial_inhibition	growth_inhibition	tumor_regression	complete_response	prolonged_survival
	not_reported					
EFFICACY_SETTINGS	prophylactic	therapeutic	unspecified			
EVIDENCE_TARGETS	immunizing_peptide	epitope	pool			
IMMUNOSUPPRESSANT_CLASSES	corticosteroid	chemotherapy	calcineurin_inhibitor	mtor_inhibitor	antimetabolite	anti_tnf
	other_immunosuppressant	unspecified				
IMMUNOSUPPRESSION_TIMING	pre_vaccine	during_priming	during_boost	throughout	post_vaccine	unspecified
LAYER2_ASSAYS	elispot	tetramer	ics	flow_cytometry	tcr_reporter	
LAYER2_OUTCOMES	immunogenic	not_immunogenic	pre_existing	not_assessed		
MEASUREMENT_TIERS	predicted	reported	validated			
MEASUREMENT_UNITS	nM	pct_rank	unknown			
MHC_CLASSES	I	II				
OUTCOMES	immunogenic	not_immunogenic	pre_existing	not_assessed	positive	negative
					presented	
PROVENANCE_KINDS	table	figure	prose	lookup		
RESPONSE_GRADES	negative	low	moderate	high	very_high	
RESPONSE_MAGNITUDE_UNITS	sfc_per_1e6	percent_of_parent	stimulation_index	unknown		
SPECIES	human	mouse	rat	non_human_primate	other	

SURVIVAL_ENDPOINTS

rfsoslandmark_rfsdfspfsefsttrother

SURVIVAL_TIME_UNITS

monthsweeksdaysyears

TIMEPOINT_PHASE

pre_vaccineon_treatmentpost_vaccinememoryunspecified

VACCINE_PLATFORMS

dnarnasynthetic_long_peptideshort_peptidedendritic_cellviral_vectorotherunspecified

VARIANT_TYPES

missenseframeshiftinframe_indelstop_losssplicefusionother